Week 4

Version Control and Organisational Coding Standards

NMTAFE | Cert IV Data Science and AI

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Professional Skills for ML and AI Workflows

Learning Objectives

- Understand core version control concepts and their relevance to AI/ML projects
- Apply git fundamentals for tracking changes and collaborating on scripts
- Implement organisational standards for script documentation and repository structure
- Connect version control skills to workplace expectations and industry best practices

Session Agenda

1. Why version control matters in ML and industry settings
2. Git basics; repositories, commits, branches, and collaboration
3. Hands-on activity; core git commands for project tracking
4. Organisational coding standards; documentation and repo structure
5. Real-world industry cases and lessons learned
6. Assessment preparation and Q&A

Why Version Control Matters

- Machine learning projects often involve large teams, frequent code changes, and strict audit requirements
- Version control systems such as git provide a reliable way to track every change, manage collaboration, and ensure reproducibility
- Industry standards require well-documented audit trails for quality assurance, licensing, and responsible AI practices

Core Concepts; Version Control

- Repository (repo); A central place to store code, scripts, and documentation
- Commit; A saved snapshot of your project at a specific point in time
- Branch; An isolated environment for separate features or experiments
- Merge; Combining different changes or branches back into a single main line
- Log; A history of all changes, for accountability and troubleshooting

Getting Started with Git

- Initiate a repository; `git init`
- Stage changes; `git add <file>`
- Save changes; `git commit -m "description"`
- View history; `git log`
- Create a branch; `git branch <feature_name>`
- Switch branches; `git checkout <branch>`

All code should be tracked in a repository for transparency and collaboration

Hands-on Example; Simple Git Workflow

1. Open your terminal and navigate to your project directory
2. `git init` ; Start a new repo
3. `git add script.py` ; Stage your Python script for tracking
4. `git commit -m "Initial commit; added script.py"`
5. Modify your script and repeat `git add` and `git commit`
6. `git log` ; Review all your changes with timestamps and messages

Practice these steps with a demo script today

Organisational Coding Standards

- Workplace scripts need clear documentation and standard structure for easy collaboration
- Standards may include
  - Header comment with author, date, purpose
  - Inline comments explaining logic
  - Clear function and variable names
  - Consistent formatting and style guide (PEP8 in Python)
- Repository structure should separate code, data, and documentation for clarity

Example; Well-Documented Script and Repo

```
# Filename; data_cleaning.py
# Author; Taylor Smith
# Date; 2024-06-05
# Purpose; Preprocess raw survey data for model training

import pandas as pd

def clean_data(df):
    # Drop missing values
    df = df.dropna()
    # Standardise column names
    df.columns = [col.lower() for col in df.columns]
    return df
```

Sample repo structure

```
project-root/
  ├── data/
  ├── scripts/
```

Practical Activity

- Create a new directory for your mini-project
- Initialise git and create a README.md
- Write and document a simple Python script
- Track your changes using `git add` and `git commit`
- Organise files into code, data, and docs folders
- Commit each change with a clear message explaining what was done

Industry Case Study; When Version Control Fails

- Several ML incidents have occurred when models were retrained using the wrong data or scripts due to missing version control
- Example; A financial institution released predictions based on an old model, costing millions in losses. Root cause; No version tracking in the ML workflow
- Lesson; Always use version control to know exactly which code and data were used to create each model

Assessment Preparation

- Mastering git and documentation skills supports the upcoming PyTorch Fundamentals Project
- Good version control practice helps with portfolio assessments and technical demonstrations
- Review all git workflows used today; these will be required as evidence in your assessment submissions

Summary and Further Learning

- Version control and documentation are essential for robust, audit-ready ML workflows
- Build the habit of clear commits, documentation, and structured repositories from day one
- Further Learning; Pro Git book (free online); Git docs at https://git-scm.com/book/en/v2
- Practice using git for every coding assignment in this course

Next Steps

- Review today’s practical exercise and repeat until confident
- Prepare homework scripts and document every change in git
- Preview next week; Introduction to PyTorch and tensor operations
- Contact instructor (in-class or by appointment) with any questions